

5.8 q_i w dV tekočine blizu kritične točke za CO_2

$$T = 25^\circ\text{C} \quad M = 44 \frac{\text{kg}}{\text{kmol}}$$

$$p_c = 73 \text{ bar}$$

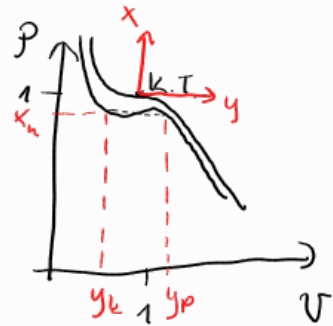
$$V_m^c = 0,095 \frac{\text{m}^3}{\text{kmol}}$$

$$\frac{T_c = 31^\circ\text{C}}{q_i = ?}$$

c.c. enačba:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

$$\frac{mq_i}{T(V - V_k)} = \frac{dp_n(T)}{dT}$$



$$\left(p + \frac{3}{v^2}\right)(3v-1) = 8T$$

$$\text{Taylor: } \frac{1}{1+u} \approx 1 - u + u^2 - \dots$$

$$\frac{1}{(1+u)^2} \approx 1 - 2u + 3u^2 - \dots$$

$$p = \frac{p}{p_c} = 1 + x$$

$$v = \frac{V_m}{V_{m,c}} = 1 + y$$

$$T = \frac{T}{T_c} = 1 + z$$

poenostavitev
v dV enačbo

$$x = -\frac{3}{2}y^3 + 4yz - 6yz^2 + 9y^2z - \frac{27}{2}y^3z + \dots$$

mehanično ravnovesje

$$1) x_h = x_k = x_p, 2) z_p = z_k$$

$$-\frac{3}{2}y_k^3 + 4yz - 6yz^2 = -\frac{3}{2}y_p^3 + 4yz - 6yz^2$$

$$3) \mu_p = \mu_k \rightarrow \text{Maxwellovo pravilo} \rightarrow (y_p - y_k)(1+x_k) = \int_{y_k}^{y_p} (1+x) dy$$

$$(y_p - y_k) + \int_{y_k}^{y_p} x dy = (y_p - y_k) + x_k(y_p - y_k)$$

$$\int_{y_k}^{y_p} \left(-\frac{3}{2}y^3 + 4z - 6yz^2\right) dy = -\frac{3}{8}(y_p^4 - y_k^4) + 4z(y_p - y_k) - 3z(y_p^2 - y_k^2)$$

$$-\frac{3}{8}(y_p^2 + y_k^2)(y_p + y_k) + 4z - 3z(y_p + y_k) = x_k \quad (1)$$

$$(2) \quad x_h = x_p$$

$$x_h = x_k \quad (3)$$

$$-\frac{3}{8}(y_p^2 + y_k^2)(y_p + y_k) - 3z(y_p + y_k) = -\frac{3}{2}y_p^3 - 6yz^2 \quad \left| \quad -\frac{3}{8}(y_p^2 + y_k^2)(y_p + y_k) - 3z(y_p + y_k) = -\frac{3}{2}y_k^3 - 6yz^2 \right.$$

$$\textcircled{2} \textcircled{3}.$$

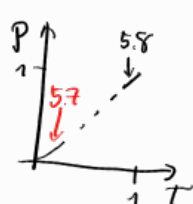
$$-\frac{3}{4}(y_p^2 + y_k^2)(y_p + y_k) = -\frac{3}{2}(y_p^3 + y_k^3)$$

$$2(y_p^3 + y_k^3) - (y_p^2 + y_k^2)(y_p + y_k) = 0$$

$$2(y_p + y_k)(y_p^2 - y_k y_p + y_k^2) - (y_p^2 + y_k^2)(y_p + y_k) = 0$$

$$(y_p + y_k)(y_p^2 - 2y_k y_p + y_k^2) = (y_p + y_k) \underbrace{(y_p - y_k)^2}_{\neq 0} = 0 \Rightarrow \underline{y_p = -y_k}$$

$$\textcircled{1} \quad x_h = 4z \rightarrow P_h - 1 = 4(\tau - 1)$$

$$P_h = 4\tau - 3$$


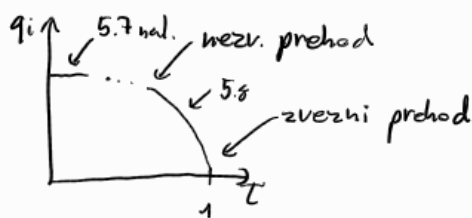
$$\textcircled{2} \quad -3zy_k = -\frac{3}{2}y_p^3 - 3y_p z$$

$$-6y_k z = \frac{3}{2}y_k^3$$

$$z = -\frac{1}{4}y_k^2, \quad y_{p,k} = \pm 2\sqrt{-z}; \quad z < 0 \rightarrow \begin{cases} v_k = 1 - 2\sqrt{1-\tau} \\ v_p = 1 + 2\sqrt{1-\tau} \end{cases}$$


$$q_i = \frac{T}{n} (v_p - v_k) \frac{dP_h}{dT} = \frac{T}{n} (v_{np} - v_{mk}) \frac{dP_h}{dT} =$$

$$= \frac{T}{n} V_m^c (v_p - v_k) \frac{P_c}{T_c} \frac{dP_h}{d\tau} = \frac{T}{n} V_m^c \cdot 4\sqrt{1-\frac{T}{T_c}} \cdot \frac{P_c}{T_c} \cdot 4 = 16 \frac{P_c V_m^c}{n} \frac{T}{T_c} \left(1 - \frac{T}{T_c}\right) = 3 \left\{ \frac{k}{kg} \right\}$$

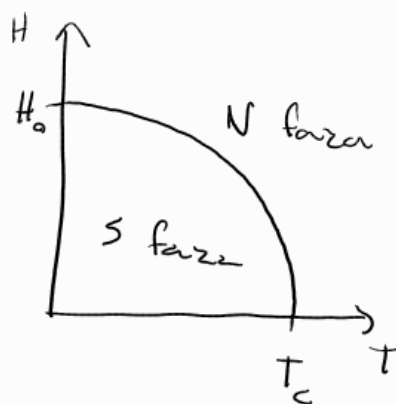


5.15) superprevodnik (M, H, T)

$$H_c(T) = H_0 \left(1 - \left(\frac{T}{T_c}\right)^2\right)$$

$$q_{sn} = ? \text{ (utajena toplota)}$$

$$C_H^S - C_H^N = ?$$



fazni diagram

$$N\text{-faza: } M=0 \rightarrow G_N = G_0(T)$$

$$S\text{-faza: } M=-H \rightarrow G_N = G_0(T, H)$$

$$Q_{SN} = (S_N - S_S)T = \left(\frac{\partial G_S}{\partial T} - \frac{\partial G_N}{\partial T} \right)_H T =$$

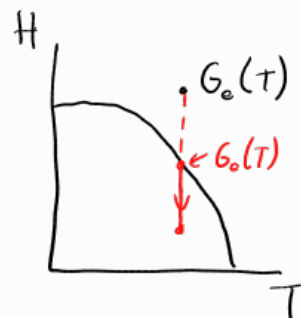
$$G = U - ST - \mu_0 V H M \quad = \frac{\partial}{\partial T} (G_S - G_N)_H T$$

$$dG = \underbrace{-S dT}_{\left(\frac{\partial G}{\partial T}\right)_H} - \underbrace{\mu_0 V M dH}_{\left(\frac{\partial G}{\partial H}\right)_T}$$

$$G_S = G_0(T) + \int_{H_c}^H \left(\frac{\partial G_S}{\partial H} \right)_T dH =$$

$$= G_0(T) - \int_{H_c}^H \mu_0 V M dH =$$

$$= G_0(T) + \mu_0 V \int_{H_c}^H H dH = G_0(T) + \frac{1}{2} \mu_0 V (H^2 - H_c^2)$$



$$Q_{SN} = \frac{1}{2} \mu_0 V T \cdot 2 \frac{dH_c}{dT} H_c = -\mu_0 V T H_0^2 \left(1 - \left(\frac{T}{T_c}\right)^2\right) \left(0 - 2 \frac{T}{T_c}\right) = Q_{sw} \uparrow$$

$$= 2 \mu_0 V T H_0^2 \frac{T}{T_c^2} \left(1 - \left(\frac{T}{T_c}\right)^2\right)$$

